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Preemptive Water Management for Oil Rim Production Using Machine Learning-Driven Data Analysis

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Abstract

The studied case addresses oil production decline in a field characterized by complex geology and challenging waterflood monitoring. A notable example is the oil rim reservoir, where water mobility significantly exceeds oil mobility due to high relative permeability. Despite low drawdown values at horizontal producers, partial aquifer support, and existing pressure maintenance facilities, oil production rates decline sharply.

The objective of the study was to reconstruct the history, pace, and future trends of formation pressure decline and to propose waterflood optimization strategies. A machine learning-based approach was applied to design these activities and enhance oil recovery.

Conventional production history was supplemented with deconvolved cross-well interference signals derived from long-term pressure records collected by permanent downhole gauges. Modern dynamic data interpretation techniques were used to assess dynamic skin factors and pressure depletion patterns. In addition, reservoir permeability and coning gradients were calculated, and open-hole logs were reinterpreted to identify geological features influencing well performance.

Machine learning models trained on historical data uncovered subtle reservoir behavior and supported decision-making in areas such as waterflood monitoring, stimulation potential, and pressure maintenance. The study made extensive use of AI for well and reservoir analysis, contributing to the development of new methodologies for extracting true mechanical skin factor in composite environments, predicting water encroachment based on skin factor dynamics, and calculating optimal drawdowns for horizontal producers.

Cross-well pressure transient analysis confirmed that the pay zone is connected to an aquifer—even during early production stages when water had not yet reached the wells. A large aquifer at the base and edges of the pay zone creates a composite system that can give the illusion of high skin around producers. This effect was carefully accounted for when calculating the true mechanical skin factor, which remained elevated—indicating potential for near-wellbore stimulation.

A decreasing skin factor over time was found to be a reliable indicator of approaching water encroachment. This insight enabled preemptive measures, such as redistributing injection volumes, to delay water breakthrough and reduce the risk of increased operating costs or expensive remediation.

The study introduced an empirical method to calculate optimal drawdown pressure values for each well individually and recommended redirecting injection efforts toward zones with more rapid pressure decline. Most of the proposed activities—including adjustments to injection and production targets, as well as targeted reservoir stimulations—were successfully implemented in the field. The observed production increase aligned closely with the study's forecasts, confirming the effectiveness of the proposed approach.

Introduction

The development of the oil rim of the Field S by horizontal wells (Fig.1), has faced significant challenges characterized by increasing water cut and a sharp oil production decline. Despite water shut-offs and other operations to mitigate production decline, such as converting production wells into injectors, the downward trend in well production persists, indicating unaccounted factors impacting performance that are not captured by standard development and monitoring techniques.

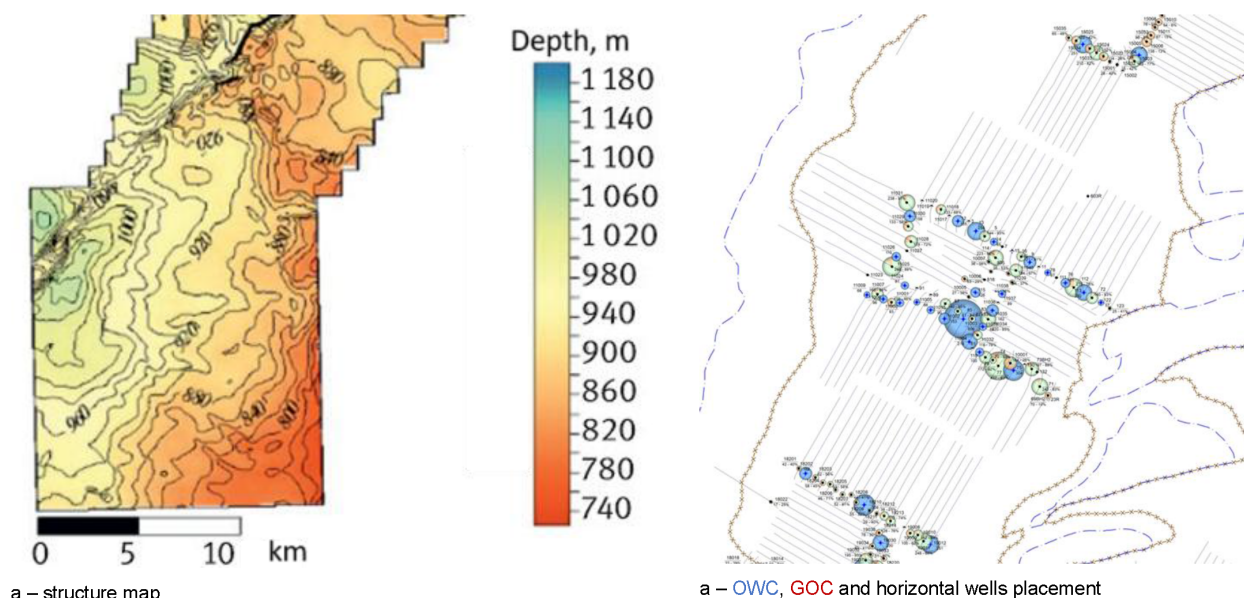


Figure 1—The oil rim of the Field S.

Further complications arise from limited well performance monitoring capabilities. The use of horizontal and multilateral wells restricts the acquisition of high-quality dynamic reservoir data, particularly regarding pressure behavior and cross-well interference. Traditional well testing techniques with well shut-ins for a pressure build-up (PBU) are often inapplicable or ineffective due to relatively short time of the test, leading to significant geological and geophysical uncertainties in reservoir management.

Historically, oil field development strategies relied heavily on conventional production data interpretation and pressure transient analysis. However, for complex reservoirs, including high-viscosity oils and low-permeability formations and thin oil rims, traditional methods of surveillance - such as standard well testing and well logging becomes misleading. geostatistical modeling face huge uncertainty and production simulation on 3D dynamic models have shown limited predictive applicability. These approaches lack sensitivity to rapidly changing reservoir conditions and fail to provide timely insights into field production decline.

Field Challenges

The asset is complex from an analytical perspective, due to several factors described below:

- Limited geological information about reservoir connectivity and unclear expectations of well interference;
- Unclear producing horizontal wellbore length because of high cost of production logging (PLT) in the wells of the field
- Declining oil productivity and unclear well skin and its changes vs time;
- High oil viscosity combined with high water mobility in the reservoir;
- Water breakthroughs in wells accompanied by increasing reservoir pressure.

The integrated reservoir study has been performed to decrease the uncertainty, increase reservoir understanding, provide better production forecast and result in the list of the most promising production enhancement operations.

Study Goals and Objectives

The engineers were tasked with achieving several key goals related to Field S. The primary goal was to assist the operator company in gaining a comprehensive understanding of the challenges affecting field performance. Another important goal was to develop potential solutions to mitigate these issues and improve overall field efficiency. A critical focus area was the ineffective waterflooding system, which had failed to deliver the expected production support, while wells were experiencing rapid water breakthrough. Ultimately, the main goal was to enhance the economic efficiency of the field development.

A set of objectives was established to provide the most effective assistance to the client:

1. To identify the primary causes of oil production decline, increased water cut, and low efficiency of the reservoir pressure maintenance system.
2. To assess detailed information on reservoir properties, geological characteristics, and cross-well interactions to support comprehensive field development strategies.
3. To develop actionable proposals aimed at enhancing the oil recovery factor and increasing the total oil flow rate.
4. To demonstrate that advanced analytical algorithms can reduce development costs and improve reservoir management under conditions of uncertainty, high risk, and limited well testing opportunities.

Method

A machine learning-based approach on base of multiwell deconvolution algorithms (MDCV) was applied to quantitatively assess cross-well interference and provide information for the waterflood optimization [1-18]. Long-term permanent downhole gauges pressure records and wells production history was analyzed to reveal unit rate pressure response on the tested wells on the offset wells production or injection. Such information is valuable for waterflood optimization. Injectors with high pressure response are a source of water breakthrough and polymer treatment is required to reduce water production. Wells with low pressure impact could have behind the casing crossflows. Each pressure transient response may be interpreted by pressure transient analysis tools to assess dynamic skin factors and pressure depletion. Wells with high skin are good candidates for well stimulation, areas with low reservoir pressure require injection increase. Machine learning models trained on historical data provided production and injection targets optimization.

Multiwell Retrospective Testing (MRT) drives insights that boost production efficiency.

MRT is a data-processing technique based on the automatic matching of production and injection rates with bottom-hole pressure readings (including those from ESP pressure gauges) in both tested wells and offset wells.

MRT enables:

- Reconstruction of formation pressure and productivity index history
- Quantification of cumulative and current cross-well interference
- Assessment of transmissibility between the wells
- Prediction of formation pressure dynamics for various field development scenarios
- Evaluation of reservoir connectivity and cross-well interference

MRT is commercial service, it based on the MDCV algorithm. Multiwell deconvolution (MDCV) – is automated rate and pressure fitting, which reconstructs initial pressure for every well and unit responses for wells and cross-well intervals. MDCV should only be used for linear diffusion (vs pressure), hence before using MDCV one should analyze the data to see if linear condition is met for the time period being tested. MDCV is applicable for stepped rate / smooth BHP mode (PTA), as well as for stepped BHP / smooth rate mode (RTA). As the procedures are symmetrical, we will further illustrate MDCV for PTA mode: switching terms and formulas to RTA mode is quite simple.

The technology is based on a mathematical algorithm. In linear formation approximation the pressure response to the varying rates in the offset wells is subject to convolution equation:

$$p_n(t) = p_{i,n} + \sum_{k=1}^N \sum_{a=1}^{N_k} (q_k^{(a)} - q_k^{(a-1)}) p_{nk}^u(t - t_{ak})$$

where: $p_n(t)$ - pressure at n-th well at arbitrary moment of time t; $p_{i,n}$ - initial pressure at n-th well; $q_n^{(a)}$ - rate value of a-th transient at n-th well; $p_{nk}^u(t)$ - unit-rate transient response from k-th well onto pressure in n-well; t_{ak} - starting point of the a-th transient in k-th well; N - number of wells in the test; N_k - number of transients in k-th well. With assumption: $p_{nk}^{(-1)} = 0$ – for any well $k = 1..N$; $p_{nk}^u(\tau) = 0$ at $\tau < 0$ for any pair of wells n, $k = 1..N$.

Hence, convolution is using initial formation pressure $p_{i,n}$, unit-rate transient responses of wells and cross-well intervals p_{nk}^u and rate histories $\{q_k(t)\}_{k=1..N}$ to calculate pressure bottom-hole pressure response as function time $p_n(t)$:

$$\left\{ p_{i,n}, \left\{ p_{nk}^u(t), q_k(t) \right\}_{k=1..N} \right\} \rightarrow p_n(t)$$

The MDCV is a reverse problem to convolution and search for N^2 functions $p_{nk}^u(t)$ and N numbers $p_{i,n}$ using the historical pressure and rate records $\left\{ p_k(t), \left\{ q_k^a \right\}_{a=1..N_k} \right\}_{k=1..N}$ and provides the adjustment to the rate histories for the small mistakes $\left\{ q_k \right\}_{k=1..N_k} \rightarrow \left\{ \tilde{q}_k \right\}_{k=1..N_k}$:

$$\left\{ p_k(t), q_k(t) \right\}_{k=1..N} \rightarrow \left\{ p_{i,n}, \left\{ p_{nk}^u(t), \tilde{q}_k(t) \right\}_{k=1..N} \right\}$$

The solution of deconvolution problem is based on the minimization of the objective function:

$$E\left(\left\{p_{i,n}, p_{n,k}^u(\tau), q_n^a\right\}_{n=1..N}\right) \rightarrow \min,$$

where

$$E\left(\left\{p_{i,n}, p_{n,k}^u(\tau), q_n^a\right\}_{n=1..N}\right) = \sum_{n=1}^N \left(p_{i,n} + \sum_{k=1}^N \sum_{a=1}^{N_k} (q_k^a - q_k^{(a-1)}) p_{n,k}^u(t - t_{ak}) - p_n(t) \right)^2 + w_c \sum_{n=1}^N \sum_{k=1}^N \text{Curv}(p_{n,k}^u(\tau)) + w_q \sum_{k=1}^N \sum_{a=1}^{N_k} (q_k^a - \tilde{q}_k^a)^2$$

One of the most efficient methods in minimizing the above objective function is the hybrid of genetic and quasi-newton algorithms.

The MDCV also adjusts the rate histories for each well $\{q_k^a\}_{a=1..N_k} \rightarrow \{\tilde{q}_k^a\}_{a=1..N_k}$ to achieve the best match of the bottom hole pressure readings. The weight coefficients and control contributions from corresponding.

MDCV limitations

The main limitations of MDCV are following:

- Central well of the test must have long term bottomhole pressure record. Surface records are not applicable for the MDCV due to the impact of multiphase fluid flow in the wellbore.
- Superposition approach is applicable to linear systems, so when saturation or skin dramatically changed MDCV is performed separately for different time periods when productivity is constant.
- Pressure and rate records must be recorded long enough to have a lot of significant rate variations in order to train the model parameters as uniquely as possible.

MRT cell

The MRT procedure is performed over a group of wells (usually from 3 to 15 wells) around a specific well (called **tested well**) (Fig. 2) with long-term bottom-hole pressure history selected by reservoir engineer to look deeper into a cross-well reservoir connectivity in the area around a tested well. The wells surrounding the tested well are referred to as offset wells.

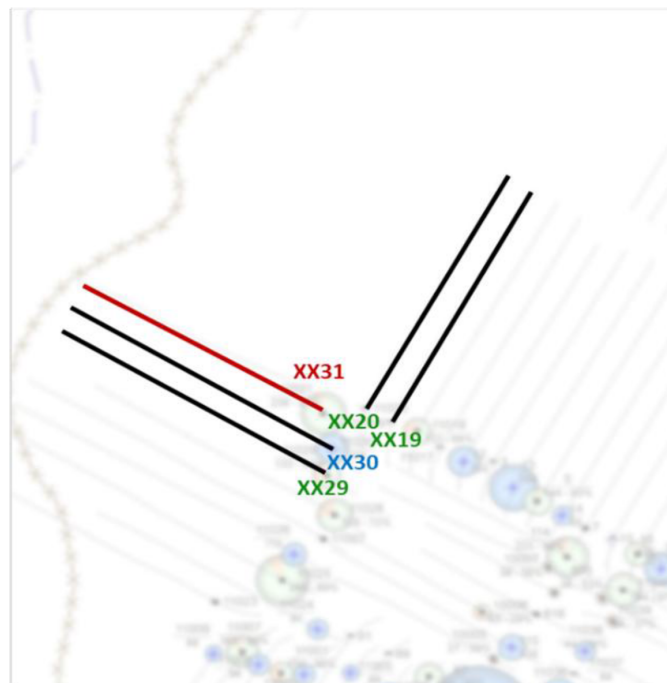


Figure 2—Cell around well XX31. Location of wells on the bubble map of cumulative production/injection.

One of the parameters calculated by the MRT algorithm is the specific cross-well connectivity. For tested wells, this parameter is drawdown transient response (DTR). It is a function of the pressure impact vs time of a unit rate production or injection of the well on its own pressure. In the case of offset wells, it is the cross-well transient response (CTR) is a pressure impact of a unit rate production or injection on the tested well pressure.

MRT validation

The MDCV solution is validated by the blind test procedure. Machine learning algorithms are used to reveal the set of DTR and CTRs on the tested data set and bottom hole pressure are calculated on the validation data set. If the cross plot of pressures demonstrates good pressure match (Fig.3) MDCV solution may be used for the forecasting.

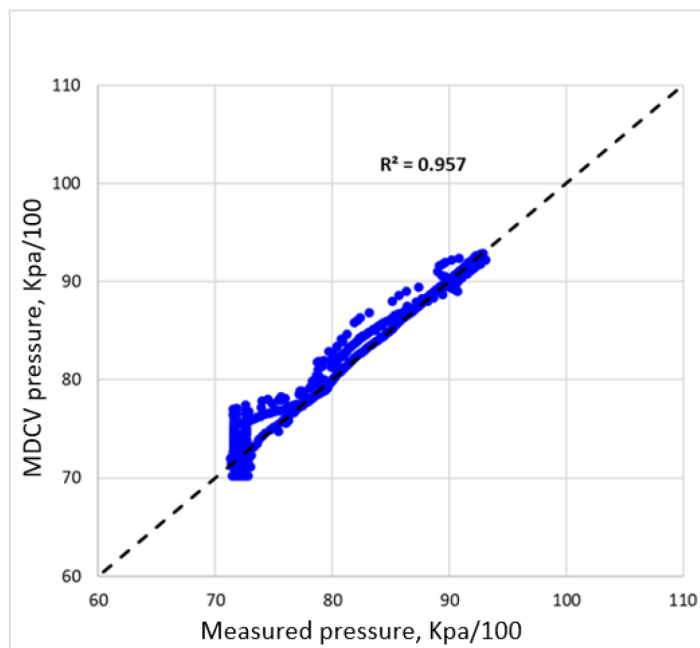


Figure 3—MDCV blind test cross plot.

MRT analysis is conducted using existing production and injection data, eliminating the need for additional data acquisition. The MRT engine is built on multi-well pressure deconvolution—a highly parallelizable decoding algorithm that runs remotely on a multi-core workstation.

The MRT workflow is supported by proprietary PolyGon software, which facilitates data processing, numerical modelling, and interpretation.

MRT novelty

It is worth saying a few words about the classical PTA and comparing it with MRT.

Pressure Transient Analysis (PTA) and Multiwell Retrospective Testing (MRT) are both diagnostic and predictive techniques used to evaluate well and reservoir performance, but they differ in scope and application. PTA focuses on the analysis of a single well's rate and pressure history using the diffusivity equation to characterize reservoir and well properties and to forecast production. The procedure typically involves diagnosing historical production and rate data, selecting an appropriate well and reservoir model, performing history matching, and generating production forecasts. In PTA, the rate and pressure responses of the well provide insight into reservoir properties, while the observed production decline also reflects the volume of drained reserves, enabling more accurate production forecasting.

MRT extends the concept of PTA to a multiwell environment and is primarily used to quantify well interference and assess opportunities for production optimization. It incorporates multiwell deconvolution in addition to the diffusivity equation to reveal interference signals between wells. The procedure includes production and rate history diagnosis, uniting transient responses from multiple wells, selecting a suitable reservoir model, evaluating well and reservoir properties, and forecasting as well as optimizing production. Unlike PTA, MRT explicitly accounts for pressure responses caused by rate changes in offset wells. Multiwell deconvolution reveals the interference history and transient responses, and their interpretation allows the determination of reservoir transmissibility and identification of potential measures for production enhancement.

Thus, while PTA is well suited for analyzing individual well performance and decline trends, MRT provides a broader, field-scale perspective by integrating the dynamic interaction between wells (Table 1). Both rely on the same physical principles of fluid flow in porous media but differ in analytical complexity and the range of insights they deliver for reservoir management and development planning.

Table 1—Comparison of PTA/RTA vs MRT results.

PTA/RTA results	MRT results
• Formation pressure • Productivity index • Well skin-factor • Well drainage area and reserves (if there are no gas cap / aquifer) • Formation transmissibility (kh/() -> permeability • Presence of no flow or constant pressure boundaries due to geology or interference (without ability to separate them)	• Formation pressure dynamic • Productivity index dynamic • Well skin-factor dynamic • Interference-free well drainage area and reserves • Formation transmissibility (kh/(μ) -> permeability • Presence of geological boundaries and quantitative interference analysis for production optimization • Quantitative history of aquifer support and each of the wells production/injection

Workflow

The workflow begins with the oil, gas, and water production history recordings, alongside injection history of injectors. This data is then synchronized with pressure records obtained through permanent downhole gauges to ensure consistency across the dataset. To facilitate accurate analysis, all data are recalculated to a common reference depth, typically corresponding to the oil-water contact (OWC), although other reference depths are also acceptable. The recalculations incorporate well sketch, phase-specific flow rates, and PVT properties of each phase.

Following data collection and initial processing, all raw data are imported into PolyGon software, which provides a comprehensive suite of preprocessing tools. The software can identify and eliminate incorrect data entries, as well as the recalculation of flow rates for each fluid phase to reservoir conditions, considering fluid compressibility and gas saturation.

The preprocessed data are then formatted appropriately for input into the multiwell deconvolution algorithm as a part of Multiwell Retrospective Testing (MRT) service (Fig. 4).

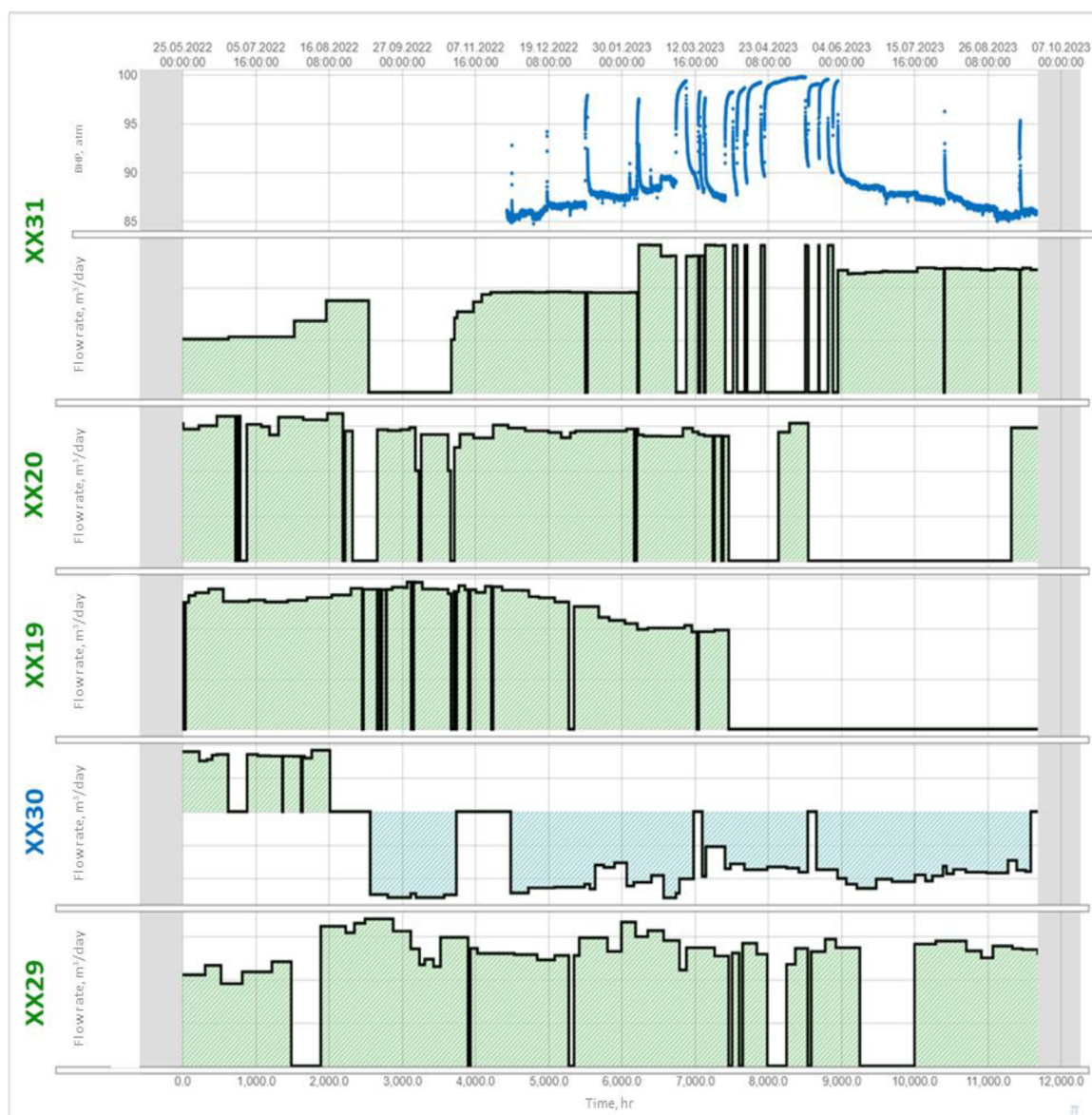


Figure 4—Production data from wells in the MRT cell surrounding the tested well X31.

This finalized dataset serves as the foundation for subsequent analysis, enabling automated, reliable evaluation of reservoir response on multiple wells production and injection. The entire workflow ensures high data quality and integrity, which are critical for obtaining accurate analysis results and provide recommendations for production enhancement activities.

Case Study

Geological and geophysical description of the field

The principal accumulations are associated with the Pokur Formation of Cenomanian age, primarily within the PK1 reservoir. The reservoir consists of unconsolidated fine-grained sandstones with lenticular and wavy bedding and abrupt changes in sand content, resulting in pronounced thin-layered heterogeneity. Alternating thin interbeds of sand and shale, poor grain sorting and weak cementation lead to low mechanical strength and complicate development operations. The trap is a structural–massive type, with a gas–condensate cap up to about 40 m thick, underlain by an oil rim of up to 20 m and basal aquifer. There are no continuous sealing barriers between the gas, oil and water zones, which promotes crossflows of gas and water into the

perforated intervals and complicates isolation. Reservoir properties are highly variable: porosity in the oil zone ranges from 27.7 % to 45.4 % with an average around 30 %, in the gas cap 23.4 % to 41.6 %, and in the oil–water transition zone approximately 34 %. Permeability varies from tens of millidarcies to over 1000 md depending on sand composition. The crude is highly viscous, with viscosity of 46–107 mPa·s and density of 0.822–0.917 g/cm³, and contains a low fraction of light components; it is mainly used as feedstock for fuel oil and specialty oils. Gas content of the reservoir rocks is estimated at 23–33 m³/t, with bubble point pressure of 6.6–10.3 MPa. The gas–condensate cap overlying the oil rim defines the field as an oil, gas and condensate accumulation. Due to the weakly cemented nature of the sandstones and absence of reliable sealing units, rapid water and gas breakthroughs are observed; contact with formation water leads to stable emulsions with viscosities up to about 2000 mPa·s, further complicating production.

Identification of significant aquifer influence

In the problematic area of the field, the formation pressure history within the drainage zone of the tested wells was reconstructed. At the same time, the current formation pressure at the time of analysis was determined. Figure 5 presents a map of the current formation pressure as calculated by the MRT method. To validate the accuracy of the pressure estimations, the results were compared with formation pressure measurements obtained from PBU test. The comparison of pressure values is depicted on the map, demonstrating a good agreement between the two datasets. Additionally, considering the initial reservoir pressure of 10,600 kPa, it can be concluded that the pressure decline during field development has been relatively modest, indicating a relatively small pressure drop over time.

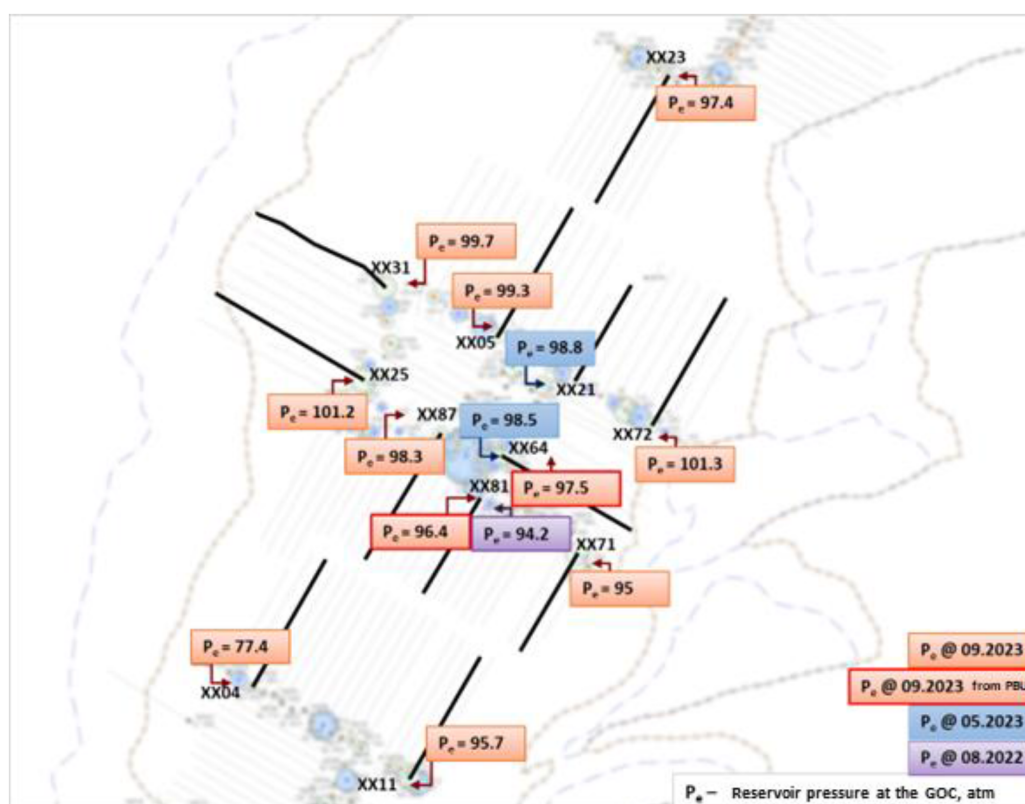


Figure 5—Formation pressure in the drainage areas of the tested wells scheme. The values are given at the analysis date, September 2023.

Using formation pressure values, the apparent reservoir transmissibility was subsequently derived. Figure 6 presents a map illustrating the distribution of apparent reservoir transmissibility. The primary distinction between this parameter and the reservoir transmissibility from the open hole logs of the reservoir lies in

its generality; it reflects the integrated transmissibility of the entire well drainage zone. As previously mentioned, the formation water mobility is high, and the underlying water is in close proximity to the wellbore. Most wells are characterized by abnormally high apparent reservoir transmissibility values according to MRT results. However, if permeability is recalculated based on the reservoir transmissibility derived from MRT, the resulting permeability values tend to be significantly overestimated and do not correspond with the permeability obtained from open-hole logging data.

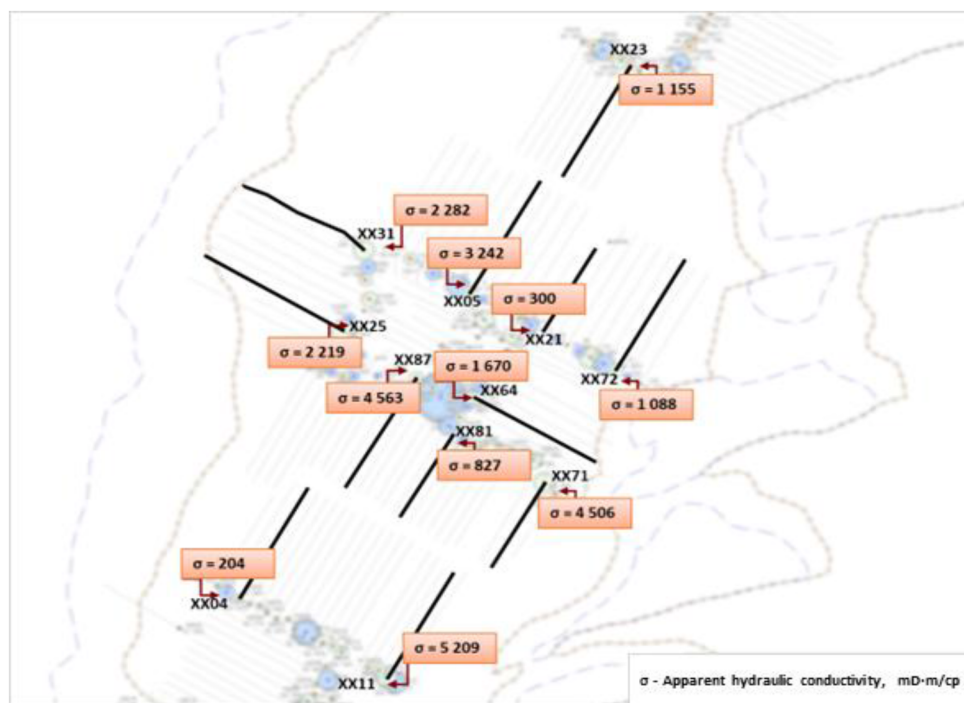


Figure 6—apparent reservoir transmissibility in the drainage areas of the tested wells scheme.

Table 2 summarizes the main parameters estimated and validated during data analysis. As observed, both parameters like reservoir transmissibility and permeability are significantly overestimated when using the reservoir transmissibility derived from MRT as the reference value. It is important to note that, despite this overestimation, the results provide valuable insights into the overall functioning of the reservoir system in Field S, since the estimated apparent reservoir transmissibility reflects a critical aspect of reservoir behavior. In particular, this forms hypotheses regarding the strong influence of the underlying aquifer and water breakthrough into the wellbore. This serves as a compelling argument in favor of adjusting the production targets for wells accordingly.

Table 2—Reservoir properties around wells.

Wells	XX23	XX31	XX05	XX21	XX72	XX64	XX25	XX87	XX81	XX71	XX04	XX11
Apparent reservoir transmissibility, mD-m/cp	1,155	2,282	3,242	300	1,088	1,670	2 219	4,563	827.3	4,560	351	5 209
Reservoir transmissibility of oil-saturated pay zone from log, mD-m/cp	316	264.5	304	270	166.4	321.4	208	255	268	342	204	140

Wells	XX23	XX31	XX05	XX21	XX72	XX64	XX25	XX87	XX81	XX71	XX04	XX11
Permeability from open-hole logging, mD	1,493.9	1,750.1	1,436.9	1,276.4	917.8	1,329.4	1 376.6	1,205	1,269.6	1,617.3	1,659.1	925.5
Permeability of oil-saturated pay zone, mD	5,459	15,100	15,323	1,418	5,999	6,906	14,683	21,566	3,910	21,557	966.5	34,471
Permeability of oil and water saturated model, mD	204	164	386	31	87	170	160	466	98	501	16.3	534
Water-saturated pay zone, m	16	40	24	28	36	28	40	28	24	26	36	28
Gas-saturated pay zone, m	24	4	18	14	10	16	4	14	24	18	8	12

The combination of factors, including a low rate of formation pressure decline, high apparent reservoir transmissibility values, and the proximity of the aquifer, indicates a dominant influence of aquifer on wells response. Most horizontal wells produce formation water from the water-saturated portion of the reservoir. In this context, the influence of the reservoir pressure maintenance system is considered negligible in terms of its impact. For example, well XX04, located in the southwest of Field S, exhibits a significantly lower formation pressure within its drained zone compared to the average formation pressure of around wells. Additionally, its apparent reservoir transmissibility is several times lower than that of other wells. This suggests that the well primarily drains the oil-saturated portion of the reservoir without substantial involvement of the aquifer.

Numerical estimation of cross-well interference

Analysis revealed that wells with high water cut (above 80%) exhibit lower values of self-interference compared to wells with low and medium water cut values. This observation can be interpreted as indicative of pressure compensation driven by the aquifer in wells with high water cut. Consequently, this pressure compensation leads to an underestimation of the self-interference values, further corroborating the previously discussed conclusion regarding the significant influence of the aquifer on the wells production within Field S.

Table 3 summarizes the water cut values and the average DTR for the tested wells within the analyzed cells.

Table 3—Comparison of water cut to average specific influence values in wells.

Wells	XX23	XX31	XX05	XX21	XX72	XX64	XX25	XX87	XX81	XX71	XX04	XX11
Water cut values of tested wells, %	20	90	90	70	95	60	90	92	25	93	40	88
DTR, kPa/m ³	5.1	1.0	0.1	10.1	4.1	2.5	2.0	1.0	4.1	1.1	28.3	2.5

It would be incorrect to characterize the waterflooding in this field as ineffective; however, the current configuration requires optimization and ongoing monitoring. The presence of weakly consolidated sandstones and high-viscosity oil, in conjunction with elevated injection rates, can lead to two adverse phenomena. First, the formation may be forced downward by the pressure exerted by the high-viscosity oil front. Second, water breakthroughs may occur in the producing wells via channels created by water from the injection wells. In such scenarios, it is essential to implement restrictions on the permissible pressure

drawdown and the rate of well drawdown increase. These measures are critical to prevent coning and water breakthroughs from the injection wells, which can significantly impair operational efficiency.

For example, in the case of well XX21, it is possible to assess the occurrence of water breakthrough from the nearest injection well. The Figure 7 presents the history of the well's downhole pressure and fluid flow rate. In mid-2023, the well was shut in, and during this period, a pressure surge was recorded, indicative of water breakthrough from the injection well. Additionally, an increase in reservoir pressure was observed following the shut-in. Such rapid changes in reservoir pressure are not attributable to aquifer influence; rather, they suggest a strong impact from the surrounding injectors.

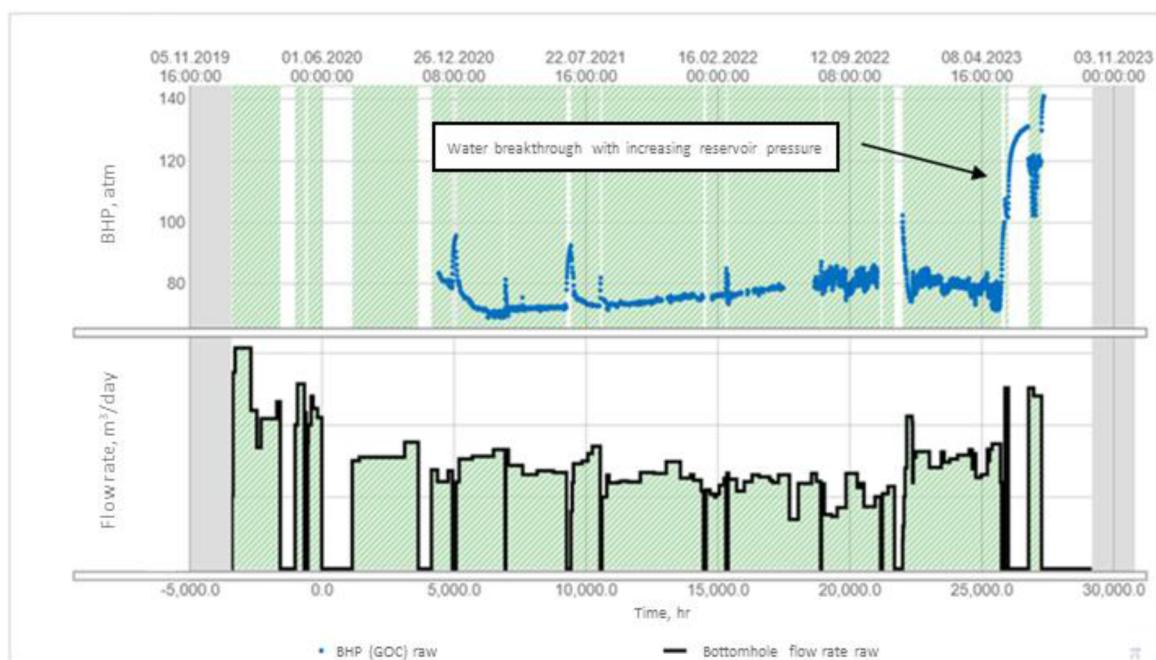


Figure 7—Pressure and flow rate of well XX21.

As previously noted, a key feature of this reservoir is that, under these conditions, water exhibits significantly higher mobility compared to oil. Therefore, the formation of flow channels extending from the injection wells to the production wells is an expected phenomenon. In most cases, water bypasses oil rather than displacing it directly. When the water front manages to influence the oil accumulation within the reservoir, it can lead to the formation of bypassed zones or drainage channels, facilitated by the rock's friability and the high viscosity of the oil. This process can result in inefficient oil recovery and operational challenges related to water production.

Water breakthrough observed at 2023 with the increase of reservoir pressure.

The aquifer water coning process was simulated using a sector numerical model implemented in PolyGon software. Figure 8 depicts the schematic of the numerical model. A two-phase reservoir model, incorporating the petrophysical characteristics of the targeted reservoir, was constructed. The horizontal section of the well penetrated solely the oil-saturated zone of the reservoir.

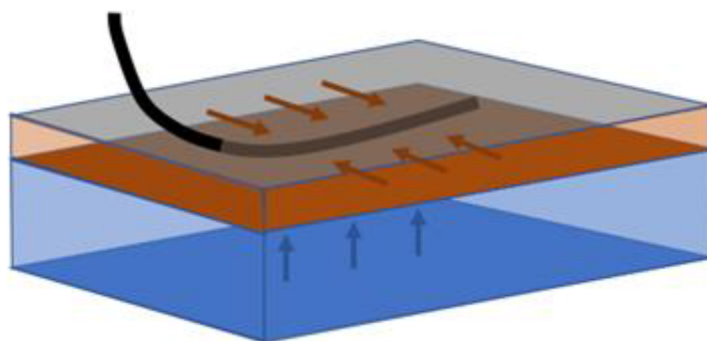


Figure 8—Schematic of the numerical model.

The constructed model demonstrates a strong correspondence with the MRT results, as evidenced by the comparison of diagnostic plots on a bilogarithmic scale, shown in Figure 9. The dotted line represents the well-formation system model obtained through sector numerical simulation. The blue and red curves correspond to the DTR of the tested wells and the derivative of the DTR, respectively. The values of the skin factor and reservoir transmissibility derived from the model sufficiently characterize the average well within the sample and effectively reflect the reservoir properties of the studied area of Field S.

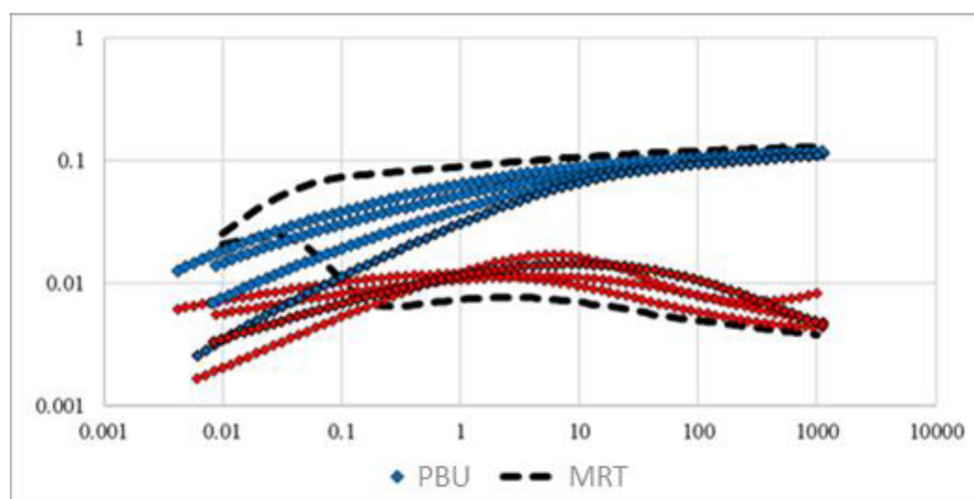


Figure 9—Schematic of the numerical model.

The results of the modeling are presented as maps of the pressure field distribution at the top and bottom layers of the reservoir, shown in Figure 10. As observed, the pressure within the reservoir is significantly disturbed in the water-saturated zone, whereas the cone of drawdown in the oil-saturated zone has primarily propagated only in the vicinity of the wellbore during the simulation period. This suggests that the primary source of reservoir pressure maintenance is the aquifer, which compensates for the pressure decline. Consequently, the unsaturated thickness of the reservoir is produced to a lesser extent. The key factors explaining this phenomenon are the high - water mobility and the improper underbalance during production.

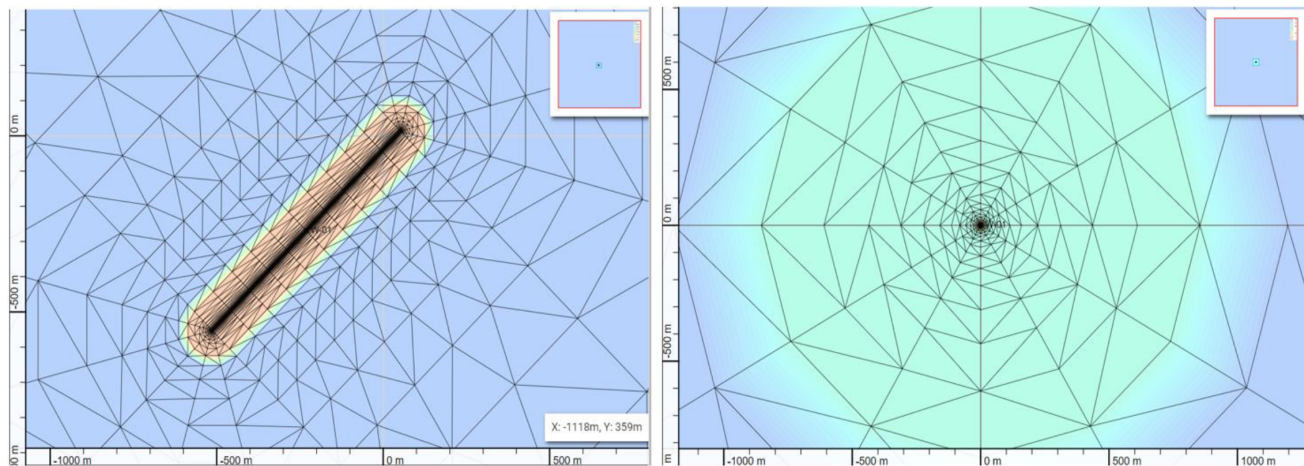


Figure 10—Pressure drawdown cone in oil-saturated zone (left) and in water-saturated zone (right).

It is noteworthy that conclusions regarding the strong influence of the aquifer can also be derived from the interpretation of standard PTA. For example, the history of well XX31 includes periods of shut-in with downhole pressure monitoring. Figure 11 presents the historical data of downhole pressure and flow rate in the upper part of the figure, while the lower part displays diagnostic plots of PBU on a bilogarithmic scale. The initial shut-in period, represented in blue on the log-log plot, is followed by a phase during which a rapid increase in pressure is observed at the beginning of the well's history after a prolonged idle period. The rate of drawdown during this period was likely excessive, resulting in a rapid descent of the drawdown cone into the aquifer. An elevated formation pressure is also evident, which can be attributed to the support of formation energy from the aquifer. Additionally, a decrease in the skin factor was observed during the shut-in period, as clearly demonstrated by the green dots on the log-log plot. However, this effect is not associated with well intervention or improvements in reservoir properties near the wellbore.

Regarding the productivity of the analyzed wells, it should be noted that, overall, the liquid productivity index has remained relatively stable over time for most wells since their last stimulation or the most recent change in flow composition. Figure 12 presents a comparison of the productivity indexes of the tested wells. It is important to emphasize that the stability of this parameter is largely maintained by the active pull-up of the aquifer. The increase in high-water-cut fluid production with rising pressure drawdown facilitates the stability of the productivity index.

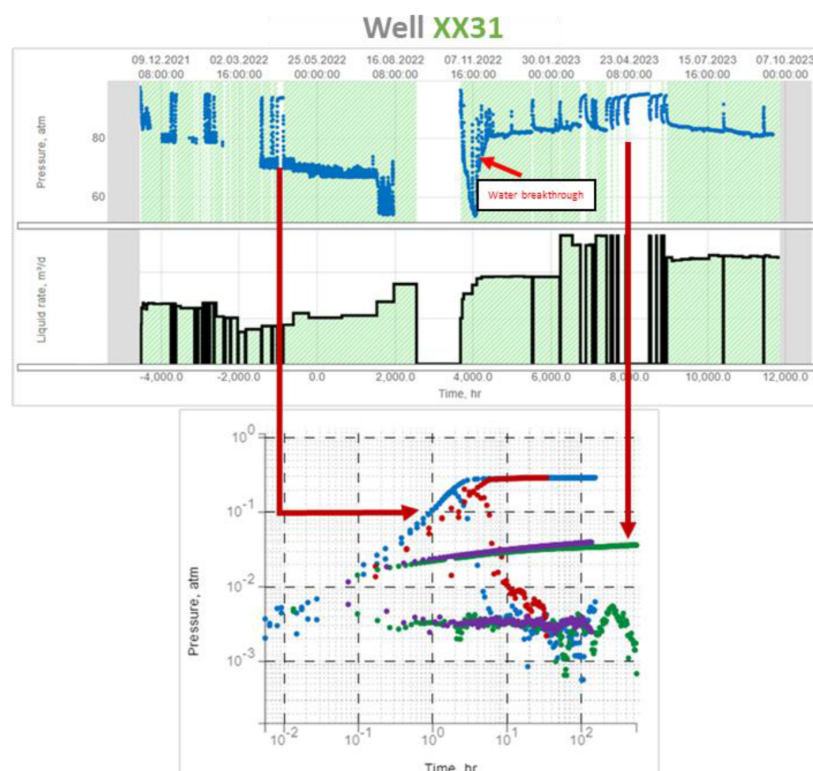


Figure 11—Decrease in the skin factor over time as a result of aquifer impact.

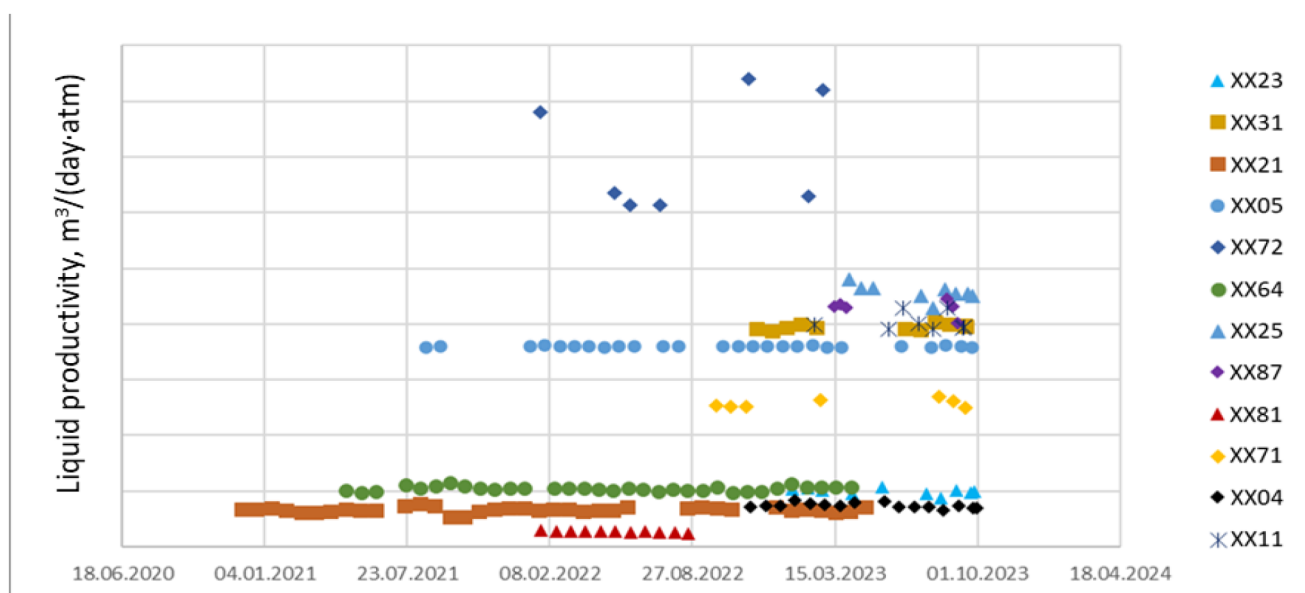


Figure 12—Comparison of liquid productivity indexes of tested wells.

Additionally, a correlation was identified when comparing well production with formation properties, which may explain the occurrence of water breakthrough in wells. It can be supposed that the maximum allowable drawdown rate should be less than 1 atm/day, and the specific pressure drawdown should not exceed 6.7 atm/m to the distance to the aquifer. Adherence to these limits can significantly reduce the risk of water breakthrough and water production, thereby enhancing the economic efficiency of the reservoir development. The specific pressure drawdown is calculated as the ratio of pressure drawdown to the effective reservoir thickness, with this value requiring individual determination for each well. Exceeding

these thresholds is associated with an increased risk of water breakthrough and water coning, which can negatively impact production profitability. It is advisable to extend this approach to other wells developing the same reservoir. [Table 3](#) summarizes the key indicators related to the risk of water breakthrough.

Table 3—Well production parameters associated with the risk of water breakthrough.

Wells	XX23	XX31	XX05	XX21	XX72	XX64	XX25	XX87	XX81	XX71	XX04	XX11
BHP of water breakthrough, kPa	7294	5399	5470	7294	7496	6270	7091	6919	3738	6119	5764	7324
	No water breakthrough			RPM water breakthrough	2 day well start-up		rapid well start-up				No water breakthrough	
Critical pressure drawdown of water breakthrough, kPa	2573	4629	4133		2269	3525	2026	2603	5551	3789	2067	2512
Distance to OWC, m	3.8	3.0	2.4–4.8	2.8	2.8	0.9–2.8	2.8	2.8	1.1–1.8	2.8	0 – 8.8	3.2–5.5
Gradient dP, kPa/m	678.71	1539.76	861.1 - 1722.1	-	810.4	1256.1 - 3920.3	719.23	931.96	3079.5 - 5044.7	1357.42	232.99	455.9 - 790.1
Apparent reservoir transmissibility by MRT, mD·m/cp	1 155	2 282	3 242	300	1 088	1 670	2 219	4 563	827	4 560	204	5 209

Well XX04 is notable among all test wells due to its distinctive characteristics. Despite the apparent proximity of its toe section to the water-oil contact (WOC) and its relatively low bottomhole pressure compared to other wells, no water breakthrough has been observed in this well. Its apparent reservoir transmissibility aligns with the estimated reservoir transmissibility of only the oil-saturated portion of the formation. Additionally, a reduction in formation pressure is observed. These features suggest that the oil-saturated thickness of the reservoir is isolated from the aquifer.

This hypothesis can be further supported by assuming that the lithotypes in the neighboring well XX41 are similar to those in well XX04. According to results from various open-hole logging methods in well XX41, shale break is identified at a depth of 1036.2–1038.2 m. In well XX04, the top of the oil-unsaturated layer is located at an absolute depth of approximately 1028 m. Consequently, it is likely that the water-saturated portion of the reservoir is overlain by this shale break. [Figure 13](#) presents the logging plot of well XX41, where the position of the shale break is indicated.

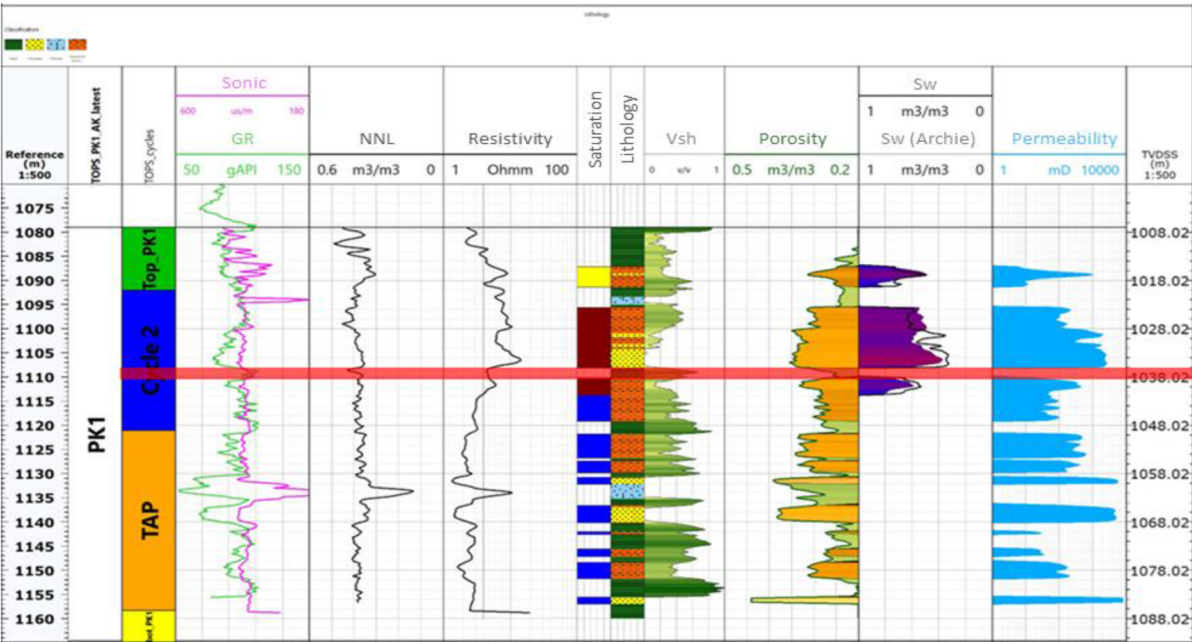


Figure 13—Open-hole logging results of well XX41.

Recommendations for Production Optimization. Based on the results of the performed analysis, several practical recommendations have been proposed to enhance well intervention and increase oil production:

The primary recommendations applicable to most of the producing well stock include conducting workover and water chut-off operations, as well as optimizing well production targets. Specifically, it is advised to adjust the bottomhole pressure while monitoring its acceptable rate of drawdown increase vs time, with well start-ups to be executed through gradual drawdown increase at a rate not exceeding 1 atm/day. For wells with water cut less than 90%, the specific drawdown at the height of the oil-water contact (OWC) should be limited to 6.7 atm/m of distance to the water saturated part of the reservoir.

Additional localized recommendations include: for well XX54, consider a gradual reduction of the bottomhole pressure through an increase in pump frequency by 5 atm, with continuous monitoring of phase flow rates; for well XX11, consider upgrading the ESP to a higher capacity model to reduce bottomhole pressure—initially by approximately 10 atm relative to the current mode—while maintaining control over phase flow rates.

It is also recommended to perform polymer treatment on injection wells to avoid water breakthrough and improve oil displacement efficiency. A targeted approach to well conversions to injection is advised, prioritizing wells exhibiting breakthroughs in "opposite phases," such as wells with increasing gas-oil ratio (GOR) where neighboring wells are experiencing water breakthrough, or vice versa. Post-implementation, the effectiveness of injectors should be reassessed after wells production and injection targets adjustment.

Finally, it is recommended to reduce the injection pressure for well XX75 to the buffer pressure value of 40 atm in order to mitigate the risk of breakthrough in adjacent producing wells.

Table 3—an increase in total oil flow rate after the implementation of recommendations in production and injection wells.

Recommendation	Flow rate increase
Wells interventions	309 m ³ /day
Wells production and injection targets adjustment	31 m ³ /day
Total	340 m ³ /day

The field possesses the potential to enhance the actual oil recovery factor. To achieve this, it is recommended to consider the implementation of infill drilling. This approach is justified by the observation that low oil recovery in the inter-well spaces persists, even in proximity to injection wells. Since high-viscosity oil is poorly displaced by water from injection wells, additional horizontal drilling between existing well rows could significantly increase the volume of recoverable reserves.

Further investigation is also advised. Specifically, it is suggested to consider production logging tools (PLT) for low-water-cut wells. This technique enables detailed flow profile analysis and allows for the estimation of the effective wellbore length. It is especially useful in evaluating completion performance under conditions where well tests do not provide an accurate assessment of the wellbore length.

Results

• Identifying Causes of Production Decline.

Analysis revealed that high water mobility, proximity to the aquifer, and inefficient pressure maintenance system due to rapid water breakthrough are the primary causes of production decline. High water cut (above 80%) correlates with low self-interference, indicating pressure compensation by the aquifer. Water cut may be reduced by optimization of well production and well injection targets.

• Reservoir Properties Assessment

MRT reconstructed formation pressure histories and determined current pressure values (e.g., current average pressure of 10,600 kPa). Values of apparent reservoir transmissibility showed high connectivity between oil saturated zones and aquifer.

• Recovery Enhancement Recommendations

Recommendations include: limiting pressure drawdown increase in each well to 100 kPa/day and overall pressure drawdown to 670 kPa/m (where m – is distance to the oil-water contact in meters), perform water shut-off operations in the wells with clear sings of water channelings behind the casing, and optimizing well injection targets. For well XX54, a gradual 500 kPa reduction in bottomhole pressure is advised; for XX11, upgrading the ESP is recommended to increase the production.

• Economic Efficiency

Compared to the FDP forecast, the oil rate in the tested area increased by approximately 2,140 bbl/day (340 m³/day) after implementing the MRT recommendations, which corresponds to a ~30% improvement.

Conclusion

This study has met its primary objective of diagnosing the key causes of production challenges in Field S and proposing the recommendations for production optimization. By applying **Multiwell Retrospective Testing (MRT)** under conditions of limited monitoring data, the work demonstrates how uncertainties can be reduced and how recovery strategies may be strengthened in fields that are otherwise difficult to manage.

Machine learning played an important role in uncovering subtle aspects of reservoir behavior that might have remained hidden using conventional methods. By training on historical data, the models provided valuable guidance in areas such as waterflood monitoring, stimulation potential, and pressure maintenance. The implementation of the recommended production-enhancement measures yielded an oil-rate increase of approximately 30% (2,140 bbl/day, or 340 m³/day - a tangible result that underscores the practical relevance of the approach.

The analysis also highlights opportunities for further improvement. In particular, low recovery in inter-well intervals, especially near injection wells, suggests the need for infill drilling. Additional horizontal wells between existing rows could significantly expand the volume of recoverable reserves, especially in

reservoirs where high-viscosity oil is not effectively displaced by water. Furthermore, the use of Production Logging Tools (PLT) in low-water-cut wells is recommended to obtain detailed flow profiles and assess the effective wellbore length, offering insights that are not accessible through conventional testing.

Taken together, the findings suggest that MRT has the potential to evolve into standard analytical practices for the management of complex reservoirs with hard-to-recover reserves. Beyond their immediate operational benefits, these methods open new possibilities for building more resilient and adaptive recovery strategies - an outcome of clear value to both operators and the broader reservoir engineering community.

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